

The Packaging Ceiling

Can Huawei's Chiplet Strategy Actually Substitute for Restricted Front-End Manufacturing?

The dominant framing for China's semiconductor challenge focuses almost entirely on the front end: SMIC cannot access EUV lithography, so it cannot produce chips at leading-edge process nodes, so its AI accelerators lag behind Nvidia's. That framing is accurate but incomplete. It misses the back end — the packaging technologies that determine how manufactured dies are assembled into functional accelerators — and it invites a tempting but flawed inference: that China could partially offset its front-end deficit by advancing its packaging capabilities faster, substituting architectural cleverness for unavailable process technology.

Huawei's Ascend 910C is the most visible test case for that proposition. It combines two 910B dies into a single package, doubling raw compute without requiring a more advanced process node. Chinese policy commentary, and some Western analysis, has treated this approach as evidence that packaging can serve as a meaningful workaround. This memo evaluates whether it can.

The answer, grounded in the physics of chip interconnects and the documented performance of the 910C on AI training workloads, is that packaging substitution works within a narrow band. Above a certain level of integration complexity, the gains from adding more dies run directly into an interconnect bandwidth wall that domestic Chinese packaging capabilities cannot currently breach. Huawei's own subsequent chip architecture decisions are the clearest indication of where that wall sits.

Why Packaging Matters for AI Performance

The performance of an AI accelerator on training workloads is not determined by raw floating-point throughput alone. It is jointly determined by compute throughput, memory bandwidth, and — critically — the speed at which data can move between the components of a chip or between chips in a cluster. Of these three, the interconnect constraint is the one most directly shaped by packaging choices.

Transformer-based model training, which underlies every competitive large language model, involves repeated all-reduce operations in which every accelerator in a cluster must share its gradient updates with every other before the next training step. Within a multi-die chip package, the same principle applies at a smaller scale: the two dies in a chiplet design must exchange data through whatever interconnect the package provides. If that interconnect is narrower or slower than the data movement that the computation requires, the chip waits. Floating-point units sit idle. Effective throughput falls below what the raw TFLOP specification would suggest.

This is not a hypothetical concern. It is the measurable gap between the Ascend 910C's headline numbers and its real-world training performance, and it originates specifically in the packaging architecture Huawei chose — or was constrained to choose.

What the 910C Actually Does

The Ascend 910C integrates two 910B dies into a single accelerator card. Each die sits on its own silicon interposer, carrying four stacks of HBM memory. The two interposers are then connected through an organic substrate — a printed circuit board-class material rather than a continuous silicon surface. The result is a dual-die package that delivers approximately 800 TFLOPS at FP16 precision and 3.2 TB/s of aggregate HBM bandwidth, reaching roughly 80 percent of the H100's headline compute figure.¹

What that number does not capture is the cost of the organic substrate connection between the two dies. An organic substrate routes signals through copper traces embedded in polymer dielectric material. The minimum achievable trace pitch on an organic substrate is constrained by the material's electrical and mechanical properties. At packaging dimensions relevant to a large AI accelerator die, organic substrate interconnects achieve line widths of roughly 10 to 15 micrometers, compared to 1 to 2 micrometers for advanced redistribution layer designs used in silicon interposer-based packaging.² The coarser pitch limits how many I/O connections can cross the substrate between the two dies, and therefore limits the raw bandwidth available across that boundary.

Industry analysts estimated, at the time of the 910C's introduction, that the organic substrate connection between the two dies delivers 10 to 20 times less die-to-die bandwidth than Nvidia achieves through NVLink on the H100.³ NVLink provides 600 GB/s of bidirectional bandwidth between paired chips. The 910C's inter-die pathway falls far short of that figure.

The consequences are measurable. DeepSeek's 2024 R2 training study — significant because it came from a Chinese laboratory with direct experience running large-scale training jobs on both platforms — found that the 910C achieved approximately 60 percent of H100 throughput on identical large language model training workloads.⁴ The gap is attributable primarily to lower effective memory bandwidth (2 TB/s measured versus the H100's roughly 4 TB/s in training-relevant configurations) and to the overhead imposed on communication-heavy operations like all-reduce by the narrow inter-die connection.⁵ The 910C's compute utilization on training — the fraction of peak TFLOPS that actually processes useful work rather than waiting for data — is correspondingly lower than the headline specification implies.

This distinction between inference and training performance matters for policy evaluation. Inference, which runs a trained model to generate outputs, is less sensitive to inter-die bandwidth because the data movement pattern is more predictable and the gradient-exchange requirement is absent. Training, which builds frontier models from scratch and requires massive, repeated all-reduce operations across thousands of chips, amplifies every bandwidth bottleneck. A chip that performs at 80 percent of H100 for inference may perform at 60 percent for training and at substantially less than that when training is distributed across thousands of cards that each carry the organic substrate penalty.

What Chinese Packaging Firms Can Produce

China's domestic outsourced semiconductor assembly and test (OSAT) industry is capable and growing. JCET and Tongfu Microelectronics are the two largest domestic players. JCET's XDFOI 2.5D pilot line entered stable mass production supplying multi-chip packaging at the 4nm node for international clients.⁶ Tongfu Microelectronics deepened its partnership with AMD to develop CoWoS-class products and raised RMB 4.4 billion in early 2026 to expand advanced packaging capacity across wafer-level and 2.5D applications.⁷ Both companies are investing in heterogeneous integration capabilities that represent a genuine step forward from conventional wire-bond or flip-chip packaging.

These achievements matter. They represent a meaningful reduction in China's historical dependence on TSMC and the major Taiwanese OSATs for back-end services. But they do not yet replicate what TSMC's CoWoS-L delivers, and the technical distance between the two is not primarily a function of investment or intent.

TSMC's CoWoS-L, the packaging technology used in Nvidia's Blackwell-architecture accelerators, uses local silicon interconnect (LSI) bridges embedded in an organic substrate to connect multiple chiplets. The silicon bridge portions achieve interconnect densities comparable to on-chip routing; the organic portions handle longer-range signal distribution. This hybrid approach allows TSMC to assemble packages that span multiple reticle fields — the GB200, which uses CoWoS-L, occupies roughly 4 reticle fields — while maintaining the high-density inter-die connections that AI training performance requires.⁸

Fabricating a silicon bridge requires photolithography at advanced nodes. The LSI elements in TSMC's CoWoS-L are processed using the same front-end equipment used to make chips, not the coarser processes used in conventional packaging. That is precisely the equipment SMIC cannot access at leading-edge geometries. A Chinese OSAT that wished to produce true CoWoS-L-class packaging would need silicon bridge components processed at nodes that currently require EUV — or, at a minimum, capable DUVi multi-patterning at scales and yields that SMIC has not yet demonstrated. The back end and the front end are not as separable as they appear.

The Deposition Layer Problem

Advanced packaging also depends on thin-film deposition processes that are less visible in policy discussions than lithography but equally consequential for achievable interconnect density.

The micro-bumps that connect a die to an interposer, and the through-silicon vias (TSVs) that carry signals through interposer layers, require precisely controlled barrier and adhesion films to prevent copper diffusion into surrounding dielectric material. At bump pitches below roughly 40 micrometers — the range relevant for high-bandwidth AI chip packaging — the barrier layer must be conformal, pinhole-free, and controlled to within a few nanometers of target thickness. Atomic layer deposition (ALD) achieves this; physical vapor deposition (PVD) and older chemical vapor deposition (CVD) techniques do not achieve the same conformality at these geometries.

ALD tools capable of packaging-relevant applications — primarily systems from Lam Research and Applied Materials — are among the equipment categories restricted by U.S. export controls. Chinese semiconductor tooling firms, including Naura Technology and ACM Research, are developing domestic alternatives. Naura in particular has made meaningful progress in thermal ALD for barrier applications. But conformality specifications, deposition rate uniformity, and throughput at production volume remain below those of restricted foreign equipment, particularly for the most demanding barrier applications at sub-30 micrometer bump pitch.⁹ The result is a practical floor on how densely Chinese packaging houses can pack I/O connections across a die boundary — and therefore a ceiling on achievable inter-die bandwidth regardless of substrate material or interposer design.

This process-level constraint receives almost no attention in policy-oriented packaging analyses. It is, however, the kind of constraint that practitioners in thin-film deposition recognize immediately: you cannot reliably control barrier integrity at fine pitch without conformal deposition, and conformal deposition at packaging-relevant geometries requires ALD or a genuine functional equivalent. The gap in Chinese ALD tooling is thus directly connected to the gap in Chinese packaging interconnect density, which is directly connected to the inter-die bandwidth penalty visible in Ascend 910C training benchmarks.

The 950PR Pivot as Evidence

The clearest evidence that Huawei has encountered the packaging ceiling described above is not the 910C's training performance numbers. It is the architecture of the chip that follows it.

Huawei's Ascend 950PR, confirmed on SMIC's N+3 (5nm-class) process in mid-2026, is a monolithic die — a single large piece of silicon rather than a multi-die chiplet package.¹⁰ This represents a deliberate reversal of the chiplet direction that the 910C embodied. Huawei packaged 1.56 PFLOPS of FP4 compute, 112 GB of proprietary HiBL memory, and a 2 TB/s interconnect into a single die, specifically — in the words of one technical analysis — "without resorting to advanced chiplet packaging it does not have access to."¹¹

A monolithic die solves the inter-die bandwidth problem by eliminating the inter-die boundary entirely. It also reintroduces the yield problem: large dies have lower yield than small dies, because any defect anywhere on the die produces a failed part. At SMIC's N+3 yield rates, a monolithic die at the area required for the 950PR carries significant yield risk. Huawei accepted that tradeoff, which indicates that the packaging bandwidth penalty was the more binding constraint.

The 950PR's inference-first optimization — it carries FP4 support but lacks FP8, which is more relevant for training — further suggests that Huawei is currently designing around its packaging and interconnect limitations rather than through them. The monolithic design maximizes within-chip bandwidth (which is not bandwidth-constrained) while Huawei separately pursues scale-out performance through its CloudMatrix rack architecture. That architecture uses 384 Ascend 910C chips connected by optical transceivers, reaching a claimed 300 petaFLOPS BF16 at the supernode level.¹² But it consumes an estimated 500 kilowatts per supernode, compared to roughly 145 kilowatts for an Nvidia GB200 NVL72 rack at similar or better performance on training workloads.¹³ Compensating for packaging limitations at the chip level by adding more chips and optical links is possible; it is not efficient.

What Packaging Cannot Substitute For

The central claim that advanced packaging can substitute for restricted front-end manufacturing holds only for a specific class of workloads and only up to a specific integration scale. For inference on models already trained, chiplet approaches — even those with organic substrate interconnects — can deliver competitive aggregate throughput, because inference tolerates the bandwidth profile that domestic Chinese packaging can achieve. For training at frontier scale, the substitution fails on two independent grounds.

The first is the interconnect bandwidth ground established above: organic substrate die-to-die connections cannot approach the bandwidth densities that CoWoS-L silicon bridge packaging achieves, and the deposition process constraints facing Chinese OSATs prevent closing that gap through finer bump pitch alone.

The second is a scaling ground that applies at the cluster level. Frontier AI training in 2026 runs on clusters of 100,000 or more accelerators. The communication overhead in such clusters — the all-reduce operations that synchronize gradients across every chip before each training step — scales with cluster size. A chip that carries a packaging-imposed bandwidth penalty of 10 to 20 times relative to its competitors passes that penalty up the hierarchy: more inter-chip communication is required to compensate for less efficient intra-chip communication, which strains the cluster fabric, which increases the communication-to-compute ratio at scale, which further degrades effective utilization.

Nvidia's NVLink and NVSwitch architecture, combined with CoWoS-L's silicon bridge inter-die connections, are not independently significant. They are a coherent hierarchy designed so that data movement is fast at every level from within the die to across the rack. Huawei's compensating strategy — high inter-chip optical bandwidth in CloudMatrix, lower intra-chip inter-die bandwidth in the 910C — creates a system whose performance characteristics diverge from Nvidia's most sharply on the workloads that matter most for frontier model development.

Implications for Export Control Policy

The packaging dimension of China's AI chip challenge has received less sustained policy attention than the lithography dimension, partly because the relevant constraints are less legible without process-level technical background, and partly because the packaging industry is geographically distributed in ways that make control harder to visualize. Several implications follow from the analysis above.

Equipment controls need to reach packaging processes. The December 2024 Bureau of Industry and Security rule focused correctly on DUV software tools that enable multi-patterning for front-end logic. An equivalent attention to the deposition and etch equipment specifically relevant to advanced packaging — ALD systems capable of sub-30 micrometer barrier applications, TSV etch tools, and advanced copper plating systems for fine-pitch redistribution layers — would address the process-level constraints that currently limit Chinese OSAT interconnect density. These tools are not universally controlled, and their relevance to packaging performance is not always legible in general-purpose export control frameworks.

Silicon bridge fabrication is a chokepoint worth naming explicitly. The capability gap between Chinese OSATs and TSMC's CoWoS-L is not principally a question of OSAT investment, process knowledge, or management capability. It is a question of whether silicon bridge components can be fabricated at the nodes required for high-density inter-die interconnects. That question runs directly through SMIC's front-end capabilities. Closing the packaging gap therefore requires closing the front-end gap first — which is a more durable structural relationship than current policy discussion acknowledges.

Inference-oriented deployments are a genuine near-term risk. The analysis above argues that packaging substitution fails for training but is more viable for inference. That is not a reassuring finding for policy purposes. A substantial portion of the strategic risk from advanced Chinese AI systems concerns not the training of new frontier models but the deployment of capable models for applications in information operations, autonomous systems, and intelligence analysis — all of which are inference workloads. Huawei's 950PR, optimized for inference on a monolithic die at SMIC N+3, is a credible inference platform that does not depend on advanced chiplet packaging. The front-end yield and scale constraints documented in the companion compute study still apply; but the inference-specific packaging gap is narrower than the training gap.

CloudMatrix's power consumption is itself an indicator. The 500-kilowatt-per-supernode figure for CloudMatrix 384 is not incidental. It is a quantitative signature of the inefficiency that results from compensating for chip-level bandwidth limitations with system-level optical interconnects and a larger number of lower-bandwidth chips. Monitoring the power density of Chinese AI data center deployments — alongside chip production volumes and CXMT's HBM ramp — provides a real-world proxy for whether the packaging bandwidth ceiling is genuinely being raised or merely worked around.

Notes

- ¹ Huawei Cloud Developer, "A Brief Introduction to Huawei Ascend Cloud," Medium (October 21, 2025); EEWorld, "Understanding Huawei's Ascend Chip in One Article" (2025).
- ² Semiconductor Engineering, "Big Changes Ahead For Interposers And Substrates" (July 17, 2025) (noting conventional PCB technology is constrained to line widths of 20 to 30 micrometers and that RDL interposers can achieve redistribution layers as fine as 1 to 2 micrometers).
- ³ @ohlennart analysis, cited in Huawei Central, "Huawei Ascend 910C Alleged Specs Suggest It a Tough Rival to Nvidia H100" (March 13, 2025); TrendForce, "Huawei's Ascend 910C Takes on NVIDIA as China's AI Race Heats Up" (March 13, 2025).
- ⁴ Electronics.alibaba.com, "What Are Huawei GPUs? Real-World Performance & Limitations" (April 30, 2026) (citing DeepSeek's 2024 R2 training study, which found the 910C achieved 60% of H100 throughput on LLM training workloads).
- ⁵ MindVL technical paper, "Towards Efficient and Effective Training of Multimodal Large Language Models on Ascend NPUs," arXiv:2509.11662 (noting that without NVLink, the Ascend 910B relies on 64 GB/s PCIe for multi-card communication, increasing latency, and that 1,800 GB/s HBM bandwidth is lower than Nvidia's equivalent).
- ⁶ GlobalSMT, "China Is Accelerating Advanced Packaging with HBM and CoWoS Amid Tightening US Restrictions" (October 1, 2024).
- ⁷ TrendForce, "China's OSAT Giants Step Up: Tongfu Microelectronics to Raise RMB 4.4B; JCET Backs Chip Fund" (January 19, 2026).
- ⁸ Oplexa, "AI Chip Packaging Bottleneck: TSMC Crisis 2026" (April 28, 2026); Financial Content/Token Ring, "The Great Packaging Pivot: How TSMC is Doubling CoWoS Capacity to Break the AI Supply Bottleneck Through 2026" (January 1, 2026).
- ⁹ This assessment draws on published specifications for domestic Chinese ALD systems from Naura Technology Group and ACM Research compared to publicly documented process requirements for sub-30 micrometer bump pitch barrier applications in advanced packaging literature, including Synopsys, "Key Considerations for Chiplet Designs: From Idea to Product" (July 17, 2025).
- ¹⁰ Tech-Insider, "Huawei Ascend 950PR: The 1.56 PFLOP AI Chip vs Nvidia" (June 2026); TechPowerUp, "Chinese SMIC Achieves 5nm Production on N+3 Node Without EUV Tools" (December 11, 2025).
- ¹¹ Tech-Insider (June 2026).
- ¹² FiberMall, "SemiAnalysis of Huawei CloudMatrix and the 910C" (September 3, 2025); Aixia, "MWC Barcelona 2026: A Deep Dive into Huawei's AI Infrastructure Stack" (March 9, 2026).
- ¹³ FiberMall (September 3, 2025) (estimating a single CloudMatrix 384 supernode at nearly 500 kilowatts, compared to approximately 145 kilowatts for an Nvidia GB200 NVL72 rack).